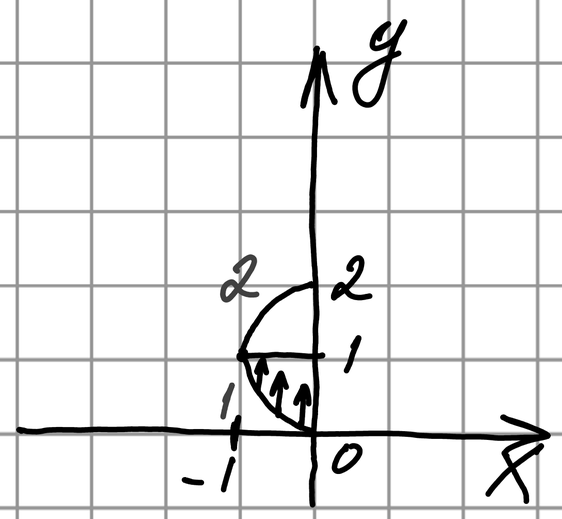
Типовик №2

1.16 Изменить порядок интегрирования:

$$\int_0^1 dy \int_{-\sqrt{y}}^{0} f\,dx + \int_1^2 dy \int_{-\sqrt{2-y}}^{0} f\,dx =$$



$$= \int_{-1}^{0} dx \int_{x^2}^{2-x^2} f\,dy$$

2.16 Найти площадь фигуры:

$G: x^2 - 2x + y^2 = 0,$

$x^2 - 4x + y^2 = 0,$

$y = 0,$

$y = \frac{x}{\sqrt{3}}.$

$\Rightarrow$

1) $x = -\sqrt{y} < 0$

$y = x^2,\ x < 0$

2) $x = -\sqrt{2-y} < 0$

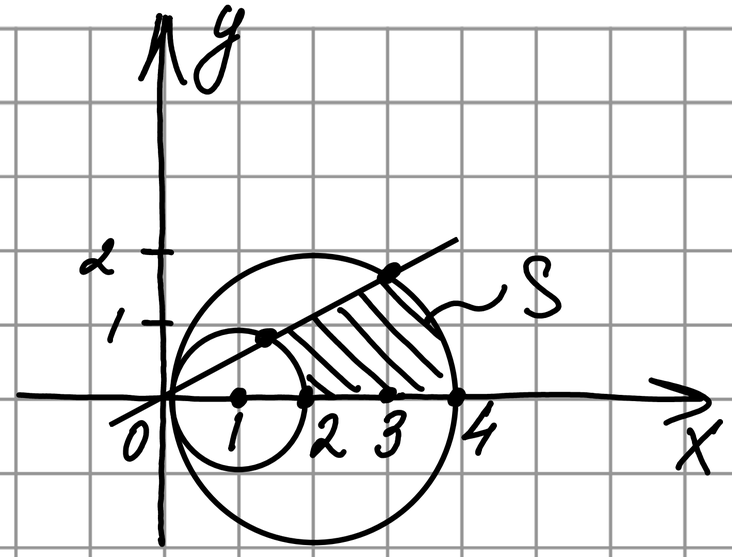
$y = 2 - x^2,\ x < 0$

⇒ $G: (x-1)^2+y^2=1,$

$(x-2)^2+y^2=4,$

$y=0,$

$y=\frac{x}{\sqrt{3}}=x\operatorname{tg}\frac{\pi}{6}$



а) $\rho^2-2\rho\cos\varphi=0\Rightarrow \rho=2\cos\varphi$

б) $\rho^2-4\rho\cos\varphi=0\Rightarrow \rho=4\cos\varphi$

$$S(G)=\int_0^{\pi/6}d\varphi\int_{2\cos\varphi}^{4\cos\varphi}\rho\,d\rho=\frac{1}{2}\int_0^{\pi/6}\left(\rho^2\Big|_{2\cos\varphi}^{4\cos\varphi}\right)\cdot d\varphi=\frac{1}{2}\int_0^{\pi/6}(16\cos^2\varphi-4\cos^2\varphi)d\varphi=$$

$$=6\int_0^{\pi/6}\cos^2\varphi\,d\varphi=\frac{6}{2}\int_0^{\pi/3}(1+\cos 2\varphi)\,d\varphi=3\cdot\frac{\pi}{6}+3\int_0^{\pi/6}\cos 2\varphi\,d\varphi=$$

$$=\frac{\pi}{2}+\frac{3}{2}\int_0^{\pi/3}\cos 2\varphi\,d2\varphi=\frac{\pi}{2}+\frac{3}{2}\cdot\frac{\sqrt{3}}{2}=\underline{\frac{\pi}{2}+\frac{3\sqrt{3}}{4}}$$

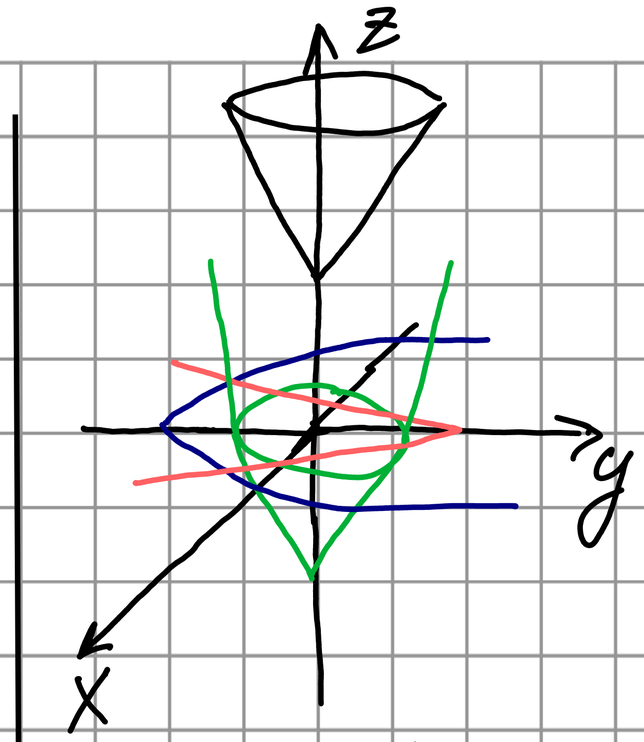
3.16 Найти объём тела:

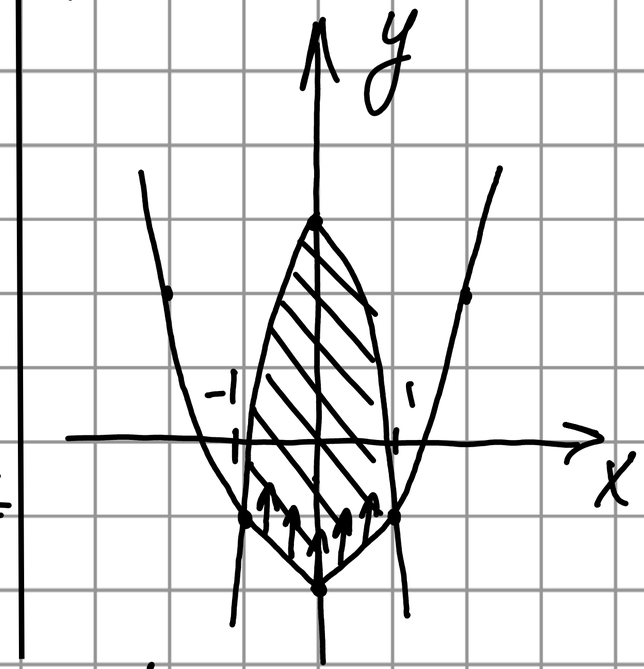
Б: $y = x^2 - 2$,

$y = -4x^2 + 3$,

$z = 2 + \sqrt{x^2+y^2} > 0$,

$z = -1 + \sqrt{x^2+y^2}$.





$$V(б) = \int_{-1}^{1} dx \int_{x^2-2}^{-4x^2+3} dy \int_{-1+\sqrt{x^2+y^2}}^{2+\sqrt{x^2+y^2}+1-\sqrt{x^2+y^2}} =$$

$$= \int_{-1}^{1} dx \int_{x^2-2}^{-4x^2+3} (2 + \cancel{\sqrt{x^2+y^2}} + 1 - \cancel{\sqrt{x^2+y^2}})\, dy =$$

$$= 3\int_{-1}^{1} (-4x^2 + 3 - x^2 + 2)\, dx = 3\int_{-1}^{1} (-5x^2 + 5)\, dx = 15\int_{-1}^{1} (1 - x^2)\, dx =$$

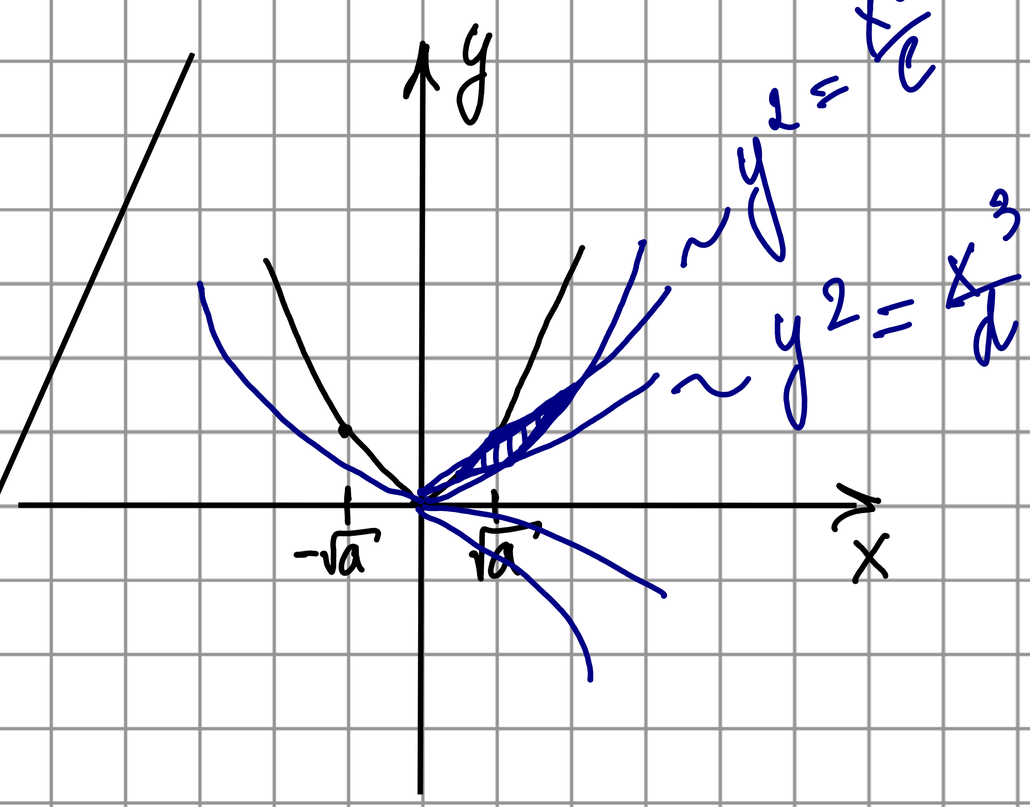
$$= 30 - 15\,\frac{x^3}{3}\Big|_{-1}^{1} = 30 - 5 \cdot 2 = \underline{\underline{20}}$$

4.16 Найти площадь фигуры, введя замену:

$G: y = \frac{x^2}{a}, y = \frac{x^2}{b}, y^2 = \frac{x^3}{c}, y^2 = \frac{x^3}{d}; \begin{matrix} 0<a<b \\ 0<c<d \end{matrix}$

Замена:

$$\begin{cases} u = \frac{x^2}{y}, \text{ где } a \le u \le b \\ v = \frac{x^3}{y^2}, \text{ где } c \le v \le d \end{cases}$$



$\frac{v}{u} = \frac{x}{y} \Rightarrow u = x \cdot \frac{v}{u} \Rightarrow x = \frac{u^2}{v}$

$y = \frac{x^2}{u} = \frac{u^3}{v^2}$

$$\begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix} = \begin{vmatrix} \frac{2u}{v} & -\frac{u^2}{v^2} \\ \frac{3u^2}{v^2} & -\frac{2u^3}{v^3} \end{vmatrix} = -\frac{4u^4}{v^4} + \frac{3u^4}{v^4} = -\frac{u^4}{v^4} \Rightarrow$$

$$\Rightarrow S(\sigma) = \int_a^b du \int_d^c \frac{u^4}{v^4}\,dv = \int_a^b -\frac{u^4\,dv}{3\cdot(c^3-d^3)} = \underbrace{\frac{1}{15}\cdot\frac{b^5-a^5}{d^3-c^3}}$$

5.16 *Вычислить объём тела:*

$$(x^2+y^2+z^2)^3 = az(x^2+y^2)^2$$

$$\begin{cases} x = r\sin\vartheta\cos\varphi \\ y = r\sin\vartheta\sin\varphi \\ z = r\cos\vartheta \end{cases} \Rightarrow \begin{vmatrix} x_r' & x_\vartheta' & x_\varphi' \\ y_r' & y_\vartheta' & y_\varphi' \\ z_r' & z_\vartheta' & z_\varphi' \end{vmatrix} = r^2\sin\vartheta$$

$$(r^2\sin^2\vartheta\cos^2\varphi + r^2\sin^2\vartheta\sin^2\varphi + r^2\cos^2\vartheta)^3 = ar\cos\vartheta\cdot$$
$$(r^2\sin^2\vartheta\cos^2\varphi + r^2\sin^2\vartheta\sin^2\varphi)^2 \Rightarrow$$

$$\Rightarrow r^6 = ar\cos\vartheta\cdot(r^2\sin^2\vartheta)^2 = ar^5\cos\vartheta\sin^4\vartheta \Rightarrow$$

$$\Rightarrow r = a\cos\vartheta\sin^4\vartheta$$

$$V = \int_0^{\pi} d\vartheta \int_0^{2\pi} d\varphi \int_0^{a\cos\vartheta\sin^4\vartheta} r^2\sin\vartheta\,dr = \frac{1}{3}\int_0^{\frac{\pi}{2}} d\vartheta \int_0^{2\pi} a^3\cos^3\vartheta\sin^{13}\vartheta\,d\varphi =$$

$$= \frac{2\pi a^3}{3} \int_0^{\frac{\pi}{2}} \cos^3\vartheta \sin^{13}\vartheta \, d\vartheta = \frac{2\pi a^3}{3} \int_0^1 \cos^2\vartheta \sin^{13}\vartheta \, d\sin\vartheta =$$

$$= \frac{2\pi a^3}{3} \int_0^1 (1-p^2) p^{13} dp = \frac{2\pi a^3}{3} \left(\int_0^1 p^{13} dp - \int_0^1 p^{15} dp \right) =$$

$$= \frac{2\pi a^3}{3} \left(\frac{1}{14} - \frac{1}{16} \right) = \frac{2\pi a^3}{3} \cdot \left(\frac{16-14}{224} \right) = \frac{\pi a^3}{3 \cdot 56} = \underline{\frac{\pi a^3}{168}}$$

6.16 Найти циркуляцию

$$\vec{a} = 4y\vec{i} - 3x\vec{j} + x\vec{k}, \quad \Gamma: \begin{cases} x = 4\cos t \\ y = 4\sin t \\ z = 4 - 4\cos t - 4\sin t = 4 - x - y \end{cases}$$

Используем криволинейный интеграл 2-го рода:

$$dx = -4\sin t \, dt; \quad dy = 4\cos t \, dt; \quad dz = 4\sin t \, dt - 4\cos t \, dt$$

$$\oint_\Gamma (4y\,dx - 3x\,dy + x\,dz) = \int_0^{2\pi} (16\sin t(-4\sin t) - 12\cos t(4\cos t) +$$

$$+ 4\cos t(4\sin t - 4\cos t)) \, dt = \int_0^{2\pi} (-64\sin^2 t - 48\cos^2 t -$$

$$- 16\cos^2 t + 8\sin 2t) \, dt =$$

$= -64 \cdot 2\pi = -128\pi$

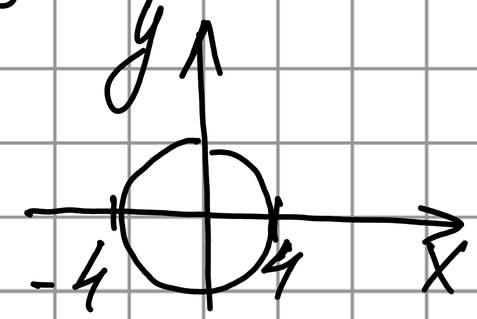
Проверим, используя ф-лу Стокса:

$$\operatorname{rot}\vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4y & -3x & x \end{vmatrix} = -3\vec{k} - 4\vec{k} - \vec{j} = -7\vec{k} - \vec{j}$$

нормаль $\vec{n}_0$ задаётся вектором $n_0\left\{\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right\}$

Имеем: $\oint (\operatorname{rot}\vec{a}, \vec{n}_0)\, dS = \frac{-8}{\sqrt{3}} \iint ds =$

$ds = \sqrt{1 + z_x'^2 + z_y'^2}\, dx\, dy = \sqrt{3}\, dx\, dy$



$$= -8 \cdot \int_0^{2\pi} d\varphi \int_0^4 r\, dr = -8 \cdot 2\pi \cdot 8 = -128\pi \checkmark$$

Значения сходятся $\Rightarrow$ модуль циркуляции $= 128\pi$

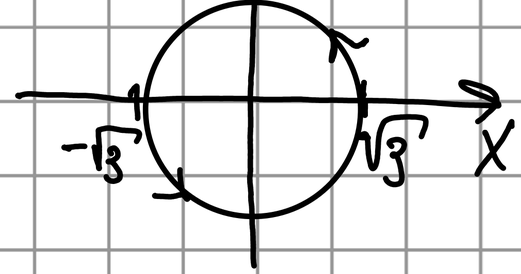
7.16 Найти циркуляцию $\vec{a}$ по Γ:

$$\vec{a} = (x-y)\vec{i} + x\vec{j} + z^2\vec{k}, \quad \Gamma: \begin{cases} x^2+y^2+z^2=4 \\ z=1 \end{cases} \Rightarrow \begin{cases} x^2+y^2=3 \\ z=1 \end{cases}$$

$$C = \oint_\Gamma (\vec{a}, \vec{b})\, dl \ominus$$

$$y = \pm\sqrt{3-x^2}$$

$$y' = \mp\frac{x}{\sqrt{3-x^2}}$$



- $dl = \sqrt{1+y_x'^2}\,dx = \sqrt{1+\frac{x^2}{3-x^2}}\,dx =$

$$= \sqrt{\frac{3-x^2+x^2}{3-x^2}}\,dx = \frac{\sqrt{3}\,dx}{|y|}$$

- $\vec{b} = \begin{cases} (-\sqrt{3-x^2}, x, 0), & y \geq 0 \\ (\sqrt{3-x^2}, x, 0), & y < 0 \end{cases}$ $\quad (-y, x, 0)$

$$\ominus \int_\Gamma \left(-y(x-y)+x^2\right)\frac{\sqrt{3}}{|y|}\,dx = \int_{\sqrt{3}}^{-\sqrt{3}} \left(-\sqrt{3-x^2}\left(x-\sqrt{3-x^2}\right)+x^2\right)\frac{\sqrt{3}}{\sqrt{3-x^2}}\,dx +$$

$$+ \int_{-\sqrt{3}}^{\sqrt{3}} \left(\sqrt{3-x^2}\left(x+\sqrt{3-x^2}\right)+x^2\right)\frac{\sqrt{3}}{\sqrt{3-x^2}}\,dx = -\sqrt{3}\int_{\sqrt{3}}^{-\sqrt{3}} \frac{-\sqrt{3-x^2}+3-x^2+x^2}{\sqrt{3-x^2}}\,dx +$$

$$+\int_{-\sqrt{3}}^{\sqrt{3}} \frac{\sqrt{3-x^2}\,(x+3)}{\sqrt{3-x^2}}\,dx = \oint_{\Gamma} dl = 2\pi\cdot\sqrt{3}\cdot\sqrt{3} = \underline{6\pi}$$

По формуле Стокса:

$$\operatorname{rot}\vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x-y & x & z^2 \end{vmatrix} = \vec{k}+\vec{k} = 2\vec{k}$$

$$\vec{n_0}\{0;0;1\}$$

$$C = \iint_S (\operatorname{rot}\vec{a}, \vec{n_0})\,dS = \iint_S 2\,ds = \underline{6\pi}$$

Циркуляция $= \underline{6\pi}$

8.16 Найти поток через указанную п-ть S:

$$\vec{a} = \underbrace{(y+6x)}_{a_x}\vec{i} + \underbrace{5(x+z)}_{a_y}\vec{j} + \underbrace{4yz}_{a_z}\vec{k}, \quad S:\begin{cases} y=x,\ y=2x,\ y=2 \\ z=x^2+y^2,\ z=0 \end{cases}$$

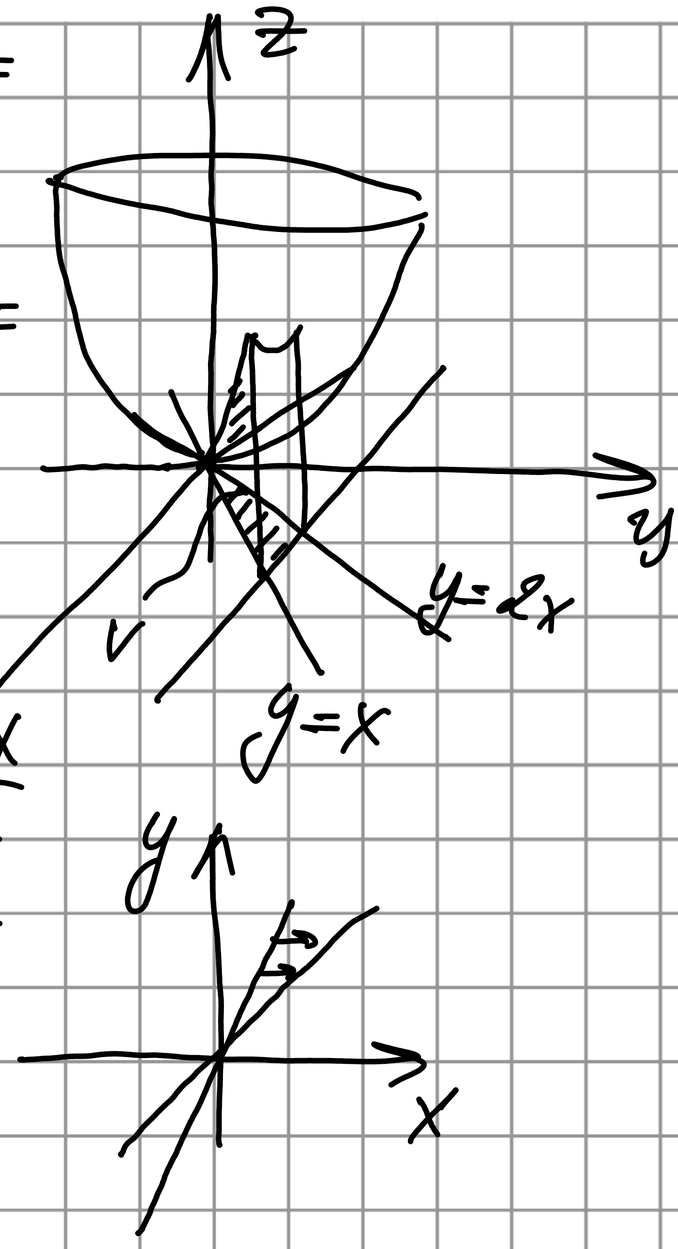
$$\operatorname{div}\vec{a} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z} = 6$$

$$P=\iiint 6\,dx\,dy\,dz = 6\iiint dx\,dy\,dz =$$

$$= \int_0^2 dy \int_{y/2}^{y} dx \int_0^{x^2+y^2} dz = \int_0^2 dy \int_{y/2}^{y} (x^2+y^2)\,dx =$$

$$= \int_0^2 \left(\frac{x^3}{3}\Big|_{y/2}^{y} + y^2 x\Big|_{y/2}^{y} \right) dy =$$

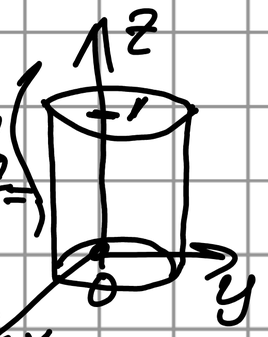
$$= \int_0^2 \left(\frac{y^3}{3} - \frac{y^3}{24} + y^3 - \frac{y^3}{2} \right) dy = \frac{5}{24}\cdot 4 = \underline{\underline{\frac{5}{6}}}$$



9.16 $\vec{a} = \underbrace{3x^2}_{a_x}\vec{i} + \underbrace{-2x^2y}_{a_y}\vec{j} + \underbrace{(2x-1)z}_{a_z}\vec{k}$, $S: \begin{cases} x^2+y^2=1 \\ z=0,\ z=1 \end{cases}$

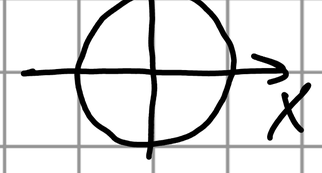
$$\operatorname{div}\vec{a} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z} = 6x - 2x^2 + 2x - 1 = -2x^2 + 8x - 1$$

$$P = \iiint \operatorname{div}\vec{a}\,dx\,dy\,dz = \int_0^{2\pi} d\varphi \int_0^1 d\rho \int_0^1 \operatorname{div}\vec{a}\cdot\rho\,dz =$$



$$= \int_0^{2\pi} d\varphi \int_0^1 d\rho \int_0^1 (-2\rho^2\cos^2\varphi + 8\rho\cos\varphi - 1)\rho\,dz =$$

$$= \int_0^{2\pi} d\varphi \int_0^1 (-2\rho^3\cos^2\varphi + 8\rho^2\cos\varphi - 2)\,d\rho =$$



$$= \int_0^{2\pi} \left(-\frac{\cos^2\varphi}{2} + \frac{8}{3}\cos\varphi - \frac{1}{2}\right) d\varphi =$$

$$= -\pi - \frac{\pi}{2} = -\frac{3\pi}{2}$$

$\begin{cases} x = 2\cos\varphi \\ y = 2\sin\varphi \\ z = z \end{cases}$

$x^2 + y^2 = 2^2 = 1$

$z = 1$

Через криволинейный второго рода:

$$P = \iint_S (3x^2\,dy\,dz - 2x^2y\,dx\,dz + (2x-1)z\,dx\,dy) =$$

$3x^2\,dy\,dz = 0$

$-2x^2y\,dx\,dz \Big|_{y=-\sqrt{1-x^2}} = -\frac{3\pi}{2}$

$(2x-1)z\,dx\,dy = 0$, т.к. $z=0$

10.1 $\vec{a} = (yz+y-1)\vec{i} + (xz+x)\vec{j} + (xy+2)\vec{k}$

Проверим $rot\,\vec{a} = 0$?

$$rot\,\vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz+y-1 & xz+x & xy+2 \end{vmatrix} = \vec{i}(x-x) + \vec{j}(y-y) + \vec{k}(z-z) = 0$$ - потенциальное поле

$$\vec{a} = grad\,p = \frac{\partial p}{\partial x}\vec{i} + \frac{\partial p}{\partial y}\vec{j} + \frac{\partial p}{\partial z}\vec{k}$$

$\frac{\partial p}{\partial x} = yz+y-1 \Rightarrow p = yzx + yx - x + c(y,z)$ ⊜

$\frac{\partial p}{\partial y} = zx + x + c_y'(y,z) = xz + x \Rightarrow c_y'(y,z) = 0 \Rightarrow c = \tilde{c}(z)$

$\frac{\partial p}{\partial z} = yx + \tilde{c}'(z) = xy + 2 \Rightarrow \tilde{c} = 2z + n$

$p = xyz + yx - x + 2z + n$

Через криволинейный интеграл: $\exists x_0=0, y_0=0, z_0=0$

$$p = p(x_0,y_0,z_0) + \int_0^x (x-1)\,dx + \int_0^y x\,dy + \int_0^z (xy+2)\,dz =$$

$$= xyz + 2z - x + xy + C$$

$$\frac{\partial p}{\partial x} = yz + y - 1 \checkmark$$

$$\frac{\partial p}{\partial y} = xz + x \checkmark$$

$$\frac{\partial p}{\partial z} = xy - 2 \checkmark$$